New Horizon Education Society's,

New Horizon Institute of Technology & Management

Kavesar, Thane-400 615

Program: AI&DS Engineering

Course File

Course: Statistics for AI&DS
Course Code: CSDLO5011
Academic Year: 2024-25

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Audit report

Course Name: Statistics for AI&DS

Semester: VI

Academic Year: 2024-25

Prepared by:

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Date	Date

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Designation:

Sign:

Date:

Institute Vision

To be a value based technical institute ensuring quality academics and fostering values to create responsible citizens of the society committed towards the development of the country.

Institute Mission

- To Create and maintain a quality technical education ecosystem.
- To promote an environ to foster ethical value systems for professional advancement of students
- To establish conception of students with firm civic sense and high moral values, committed for the development of the nation

Department Vision

To confer domain specific education enabling the students to enhance their analytical skills in the area of artificial intelligence and data science to cater to the growing needs of this industry

Department Mission

1. Endowing the students with the knowledge of the transformative power of Artificial Intelligence and data science and use it to solve real-world challenges
2. Equip the students with statistical, mathematical reasoning, machine learning, algorithm and knowledge discovery
3. Enable students to acquire the required technological knowledge and skills, thus enhancing their professional prospects in the industry

Program Outcomes

Program Outcome Code	Program Outcomes
PO1	Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4	Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO7	Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9	Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Educational Objectives and Program Specific Outcomes

Program Educational Objectives:

PEO Code	Program Specific Outcome
PEO1	Mining insightful knowledge vital for AI & Data Science domain for the industry and society
PEO2	Nurture learning and research skills to adapt and cater to this rapidly evolving domain of Artificial Intelligence and Data Science.
PEO3	Enable the students with contemporary technological skills and aptitude for entrepreneurship.

Program Specific Outcomes:

PSO Code	Program Specific Outcome
PSO1	To focus on developing analytical and computational skills relating to the Artificial Intelligence and Data Science domain and solve real world challenges.
PSO2	To inculcate knowledge of Artificial Intelligence and Data Science techniques such as data analysis, machine learning, data mining, algorithms using contemporary technologies.
PSO3	To imbibe lifelong learning skills in students essential for progression in their careers.

Syllabus

Module		Detailed Content	Hours
1		**Exploratory Data Analysis**	5
	1.1	Elements of Structured Data ,Further Reading ,Rectangular Data ,Data Frames and Indexes ,Nonrectangular Data Structures , Estimates of Location ,Mean ,Median and Robust Estimates , Estimates of Variability,Standard Deviation and Related Estimates ,Estimates Based on Percentiles , Exploring the Data Distribution ,Percentiles and Boxplots ,Frequency Tables and Histograms ,Density Plots and Estimates.	
	1.2	Exploring Binary and Categorical Data , Mode ,Expected Value, Probability ,Correlation ,Scatterplots ,Exploring Two or More Variables ,Hexagonal Binning and Contours (Plotting Numeric Versus Numerical Data) ,Two Categorical Variables ,Categorical and Numeric Data ,Visualizing Multiple Variables.	
2		**Data and Sampling Distributions**	6
	2.1	Random Sampling and Sample Bias ,Bias ,Random Selection ,Size Versus Quality,Sample Mean Versus Population Mean ,Selection Bias ,Regression to the Mean ,Sampling Distribution of a Statistic ,Central Limit Theorem ,Standard Error ,The Bootstrap ,Resampling Versus Bootstrapping .	
	2.2	Confidence Intervals ,Normal Distribution ,Standard Normal and QQ-Plots,Long-Tailed Distributions ,Student's t-Distribution ,Binomial Distribution ,Chi-Square Distribution ,F-Distribution ,Poisson and Related Distributions ,Poisson Distributions ,Exponential Distribution ,Estimating the Failure Rate ,Weibull Distribution . **Self Study :** Problems in distributions.	
3		**Statistical Experiments and Significance Testing**	8
	3.1	A/B Testing ,Hypothesis Tests ,The Null Hypothesis ,Alternative Hypothesis ,One-Way Versus Two-Way Hypothesis Tests ,Resampling ,Permutation Test ,Example: Web Stickiness,Exhaustive and Bootstrap Permutation Tests ,Permutation Tests: TheBottom Line for Data Science ,Statistical Significance and p-Values,p-Value ,Alpha ,Type 1 and Type 2 Errors	

	3.2	Data Science and p-Values , t-Tests ,Multiple Testing ,Degrees of Freedom ,ANOVA ,F-Statistic ,Two-Way ANOVA , Chi-Square Test ,Chi-Square Test: A Resampling Approach ,Chi-Square Test: Statistical Theory ,Fisher's Exact Test ,Relevance for Data Science ,Multi-Arm Bandit Algorithm ,Power and Sample Size ,Sample Size . Self Study : Testing of Hypothesis using any statistical tool	
4		**Summarizing Data**	6
	4.1	Methods Based on the Cumulative Distribution Function , The Empirical Cumulative Distribution Function ,The Survival Function ,Quantile-Quantile Plots , Histograms, Density Curves, and Stem-and-Leaf Plots , Measures of Location.	
	4.2	The Arithmetic Mean ,The Median , The Trimmed Mean , M Estimates , Comparison of Location Estimates ,Estimating Variability of Location Estimates by the Bootstrap , Measures of Dispersion , Boxplots , Exploring Relationships with Scatterplots . Self Study : using any statistical tool perform data summarization	
5		**The Analysis of Variance**	6
	5.1	The One-Way Layout, Normal Theory; the F Test ,The Problem of Multiple Comparisons , A Nonparametric Method—The Kruskal-Wallis Test ,The Two-Way Layout , Additive Parametrization , Normal Theory for the Two-Way Layout ,Randomized Block Designs , A Nonparametric Method—Friedman's Test .	
6		Linear Least Squares	8
	6.1	Simple Linear Regression, Statistical Properties of the Estimated Slope and Intercept , Assessing the Fit , Correlation and Regression , The Matrix Approach to Linear Least Squares , Statistical Properties of Least Squares Estimates , Vector-Valued Random Variables , Mean and Covariance of Least Squares Estimates , Estimation of $\sigma 2$, Residuals and Standardized Residuals , Inference about β , Multiple Linear Regression—An Example , Conditional Inference, Unconditional Inference, and the Bootstrap , Local Linear Smoothing . **Self-Study**: Create a Linear Regression model for a dataset and display the error measures, Chose a dataset with categorical data and apply linear regression model	

Text Books:

1. Bruce, Peter, and Andrew Bruce. Practical statistics for data scientists: 50 essential concepts. Reilly Media, 2017.
2. Mathematical Statistics and Data Analysis John A. Rice University of California, Berkeley,Thomson Higher Education

Reference Books:

1. Dodge, Yadolah, ed. Statistical data analysis and inference. Elsevier, 2014.
2. Ismay, Chester, and Albert Y. Kim. Statistical Inference via Data Science: A Modern Dive into R and the Tidyverse. CRC Press, 2019.
3. Milton.J. S. and Arnold. J.C., "Introduction to Probability and Statistics", Tata McGraw Hill, 4th Edition, 2007.
4. Johnson. R.A. and Gupta. C.B., "Miller and Freund's Probability and Statistics for Engineers", Pearson Education, Asia, 7th Edition, 2007.
5. A.Chandrasekaran,G.Kavitha,—Probability,Statistics,RandomProcesses andQueuingTheoryl,Dhanam Publications, 2014.

Useful Links:

1. https://www.edx.org/course/introduction-probability-science-mitx-6-041x-2
2. https://www.coursera.org/learn/statistical-inference
3. https://www.datacamp.com/community/open-courses/statistical-inference-and-data-analysis

Content Beyond Curriculum

Sr. No.	Topic	PO Mapping	Mode of Content Delivery	Date
1	**Unit-2** Concept of probability, Bayes Theorem, Binomial theorem, exponential Theorem.	PO1, PO2, PO3, PO12	Visualization and Traditional class	19/7/2024

Course Outcomes

CO	Course Outcome At the end of the course student will be able to:	Cognitive Level (Remember, Understand, Apply, Create, Analyze, Evaluate)	Hours Planned
CSDLO5011.1	Illustrate Exploratory Data Analysis	UNDERSTAND	05
CSDLO5011.2	Describe Data and Sampling Distributions	APPLY	06
CSDLO5011.3	Solve Statistical Experiments and Significance Testing	EVALUATE	08
CSDLO5011.4	Demonstrate Summarizing Data	APPLY	06
CSDLO5011.5	Interpret the Analysis of Variance	EVALUATE	06
CSDLO5011.6	Use Linear Least Squares	EVALUATE	08
		Total Hours Planned	**39**

Modes of Content Delivery

i. Traditional class
ii. Visualization
iii. Flipped Class
iv. TPS (Think Pair Share)
v. Expert Lecture
vi. Seminar
vii. Narrated PPT
viii. Online Video Lectures

Code	Content Delivery	No of Hours
i	Traditional class	20
ii	Visualization	5
iii	Flipped Class	
iv	TPS (Think Pair Share)	
v	Expert Lecture	
vi	Workshop	
vii	Online Video Lecture	
	Total No of Hours	

Lesson Plan

Lecture No.	Topic Planned	Conducted Date	Cognitive Level	Mode of content Delivery	Text Books & Reference Books	Mapped with CO:PO
1	**Unit 1: Exploratory Data Analysis:** Elements of Structured Data, Further Reading, Rectangular Data, Data Frames and Indexes, Nonrectangular Data Structures	8/7/2024 and 9/7/2024	Understand	Traditional Class + Visualization	T1,T2,R3	CSDLO5011.1 : PO1.PO2, PO12
2	Estimates of Location, Mean, Median and Robust Estimates, Estimates of Variability, Standard Deviation and Related Estimates, Estimates Based on Percentiles	10/7/2024	Understand	Traditional Class + Visualization	T1,T2,R3	CSDLO5011.1 : PO1.PO2, PO3, PO12
3	Exploring the Data Distribution, Percentiles and Boxplots, Frequency Tables and Histograms, Density Plots and Estimates.	11/7/2024	Understand	Traditional Class + Visualization	T1,T2,R3	CSDLO5011.1 : PO1.PO2, PO3, PO12
4	Exploring Binary and Categorical Data, Mode, Expected Value, Probability, Correlation, Scatterplots	12/7/2024	Understand	Traditional Class + Visualization	T1,T2,R3	CSDLO5011.1 : PO1.PO2, PO3, PO12
5	Exploring Two or More Variables, Hexagonal Binning and Contours (Plotting Numeric Versus Numerical Data), Two Categorical Variables, Categorical and Numeric Data, Visualizing Multiple Variables.	15/07/2024	Understand	Traditional Class + Visualization	T1,T2,R3	CSDLO5011.1 : PO1.PO2, PO3, PO12
6	**Unit-2: Data and Sampling Distributions:** Random Sampling and		Understand	Traditional Class +	T1,T2,R1,R2	CSDLO5011.2 : PO1.PO2,

	Sample Bias, Bias, Random Selection, Size Versus Quality	19/7/2024		Visualization		PO12
7	Sample Mean Versus Population Mean, Selection Bias, Regression to the Mean, Sampling Distribution of a Statistic	29/7/2024	Understand	Traditional Class + Visualization	T1,T2,R1,R2	CSDLO5011.2 : PO1.PO2, PO12
8	Central Limit Theorem, Standard Error, The Bootstrap, Resampling Versus Bootstrapping.	29/7/2024	Understand	Traditional Class	T1,T2,R1,R2	CSDLO5011.2 : PO1.PO2, PO3, PO12
9	Confidence Intervals, Normal Distribution, Standard Normal and QQ-Plots, Long-Tailed Distributions	31/7/2024	Apply	Traditional Class + Visualization	T1,T2,R1,R2	CSDLO5011.2 : PO1.PO2, PO3, PO12
10	Student 's t-Distribution, Binomial Distribution, Chi-Square Distribution	31/7/2024	Apply	Traditional Class	T1,T2,R1,R2	CSDLO5011.2 : PO1.PO2, PO3, PO12
11	F-Distribution, Poisson and Related Distributions, Poisson Distributions, Exponential Distribution, Estimating the Failure Rate, Weibull Distribution	1/8/2024	Apply	Traditional Class	T1,T2,R1,R2	CSDLO5011.2 : PO1.PO2, PO3, PO12
12	**Unit-3: Statistical Experiments and Significance Testing:** A/B Testing, Hypothesis Tests, The Null Hypothesis, Alternative Hypothesis	5/8/2024	Understand	Traditional Class	T1,R1,R2,R3	CSDLO5011.3 : PO1.PO2, PO3, PO12
13	One-Way Versus Two-Way Hypothesis Tests, Resampling, Permutation Test	7/8/2024	Apply	Traditional Class	T1,R1,R2,R3	CSDLO5011.3 : PO1.PO2, PO3, PO4, PO12
14	Example: Web Stickiness, Exhaustive and Bootstrap Permutation Tests	9/8/2024	Apply	Traditional Class	T1,R1,R2,R3	CSDLO5011.3 : PO1.PO2, PO3, PO4, PO12

15	Permutation Tests: The Bottom Line for Data Science, Statistical Significance and p-Values-Value, Alpha, Type 1 and Type 2 Errors	12/8/2024	Analyze	Traditional Class	T1,R1,R2,R3	CSDLO5011.3 : PO1.PO2, PO3, PO12
16	Data Science and p-Values, t-Tests, Multiple Testing, Degrees of Freedom, ANOVA, F-Statistic, Two-Way ANOVA	14/8/2024	Evaluate	Traditional Class	T1,R1,R2,R3	CSDLO5011.3 : PO1.PO2, PO3, PO4, PO12
17	Chi-Square Test ,Chi-Square Test: A Resampling Approach ,Chi-Square Test: Statistical Theory	16/8/2024	Evaluate	Traditional Class	T1,R1,R2,R3	CSDLO5011.3 : PO1.PO2, PO3, PO4, PO12
18	Fisher 's Exact Test, Relevance for Data Science	19/8/2024	Evaluate	Traditional Class	T1,R1,R2,R3	CSDLO5011.3 : PO1.PO2, PO3, PO4, PO12
19	Multi-Arm Bandit Algorithm, Power and Sample Size, Sample Size	21/8/2024	Analyze	Traditional Class	T1,R1,R2,R3	CSDLO5011.3 : PO1.PO2, PO3, PO12
20	**Unit-4: Summarizing Data** Methods Based on the Cumulative Distribution Function, The Empirical Cumulative Distribution Function, The Survival Function		Understand	Traditional Class	T2,R3,R4	CSDLO5011.4 : PO1.PO2, PO3, PO12
21	Quantile-Quantile Plots, Histograms, Density Curves,		Apply	Traditional Class + Visualization	T2,R3,R4	CSDLO5011.4 : PO1.PO2, PO3, PO4, PO12
22	Stem-and-Leaf Plots, Measures of Location		Apply	Traditional Class	T2,R3,R4	CSDLO5011.4 : PO1.PO2, PO3, PO4, PO12

23	The Arithmetic Mean, The Median, The Trimmed Mean, M Estimates		Understand	Traditional Class	T2,R3,R4	CSDLO5011.4 : PO1.PO2, PO3, PO4, PO12
24	Comparison of Location Estimates, Estimating Variability of Location Estimates by the Bootstrap		Understand	Traditional Class	T2,R3,R4	CSDLO5011.4 : PO1.PO2, PO3, PO4, PO12
25	Measures of Dispersion, Boxplots, Exploring Relationships with Scatterplots.		Apply	Traditional Class + Visualization	T2,R3,R4	CSDLO5011.4 : PO1.PO2, PO3, PO4, PO12
26	**Unit-5: The Analysis of Variance:** The One-Way Layout, Normal Theory; the F Test		Understand	Traditional Class	T1,R2,R3,R4	CSDLO5011.5 . PO2, PO3, PO4, PO12
27	The Problem of Multiple Comparisons		Evaluate	Traditional Class	T1,R2,R3,R4	CSDLO5011.5 . PO1, PO2, PO3, PO4, PO12
28	A Nonparametric Method—The Kruskal-Wallis Test		Evaluate	Traditional Class	T1,R2,R3,R4	CSDLO5011.5 . PO1, PO2, PO3, PO12
29	The Two-Way Layout		Understand	Traditional Class	T1,R2,R3,R4	CSDLO5011.5 . PO1, PO2, PO3, PO4, PO12
30	Additive Parametrization, Normal Theory for the Two-Way Layout, Randomized Block Designs		Analyze	Traditional Class	T1,R2,R3,R4	CSDLO5011.5 . PO1, PO2, PO3, PO4, PO12
31	A Nonparametric Method—Friedman 's Test.		Evaluate	Traditional Class	T1,R2,R3,R4	CSDLO5011.5 . PO1, PO2, PO3, PO4, PO12

32	**UNIT-6: Linear Least Squares** Simple Linear Regression, Statistical Properties of the Estimated Slope and Intercept, Assessing the Fit		Understand	Traditional Class	T1,T2, R2	CSDLO5011.6 . PO1, PO2, PO3, PO4, PO12
33	Correlation and Regression		Analyze	Traditional Class	T1,T2, R2	CSDLO5011.6 . PO1, PO2, PO3, PO4, PO12
34	The Matrix Approach to Linear Least Squares, Statistical Properties of Least Squares Estimates		Evaluate	Traditional Class	T1,T2, R2	CSDLO5011.6 . PO1, PO2, PO3, PO4, PO12
35	, Vector-Valued Random Variables, Mean and Covariance of Least Squares Estimates		Analyze	Traditional Class	T1,T2, R2	CSDLO5011.6 . PO1, PO2, PO3, PO4, PO12
36	Estimation of $\sigma 2$, Residuals and Standardized Residuals, Inference about β		Evaluate	Traditional Class	T1,T2, R2	CSDLO5011.6 . PO1, PO2, PO3, PO4, PO12
37	Multiple Linear Regression—An Example		Evaluate	Traditional Class	T1,T2, R2	CSDLO5011.6 . PO1, PO2, PO3, PO4, PO12
38	Conditional Inference, Unconditional Inference, and the Bootstrap		Analyze	Traditional Class	T1,T2, R2	CSDLO5011.6 . PO1, PO2, PO3, PO4, PO12
39	Local Linear Smoothing.		Analyze	Traditional Class	T1,T2, R2	CSDLO5011.6 . PO1, PO2, PO3, PO4, PO12

CO-PO Mapping

Course Code	Course Outcome At the end of the course student will be able to:	Mapped PO
CSDLO5011.1	Illustrate Exploratory Data Analysis	PO1, PO2, PO3, PO12
CSDLO5011.2	Describe Data and Sampling Distributions	PO1, PO2, PO3, PO12
CSDLO5011.3	Solve Statistical Experiments and Significance Testing	PO1, PO2, PO3, PO4, PO12
CSDLO5011.4	Demonstrate Summarizing Data	PO1, PO2, PO3, PO4, PO12
CSDLO5011.5	Interpret the Analysis of Variance	PO1, PO2, PO3, PO4, PO12
CSDLO5011.6	Use Linear Least Squares	PO1, PO2, PO3, PO4, PO12

CO-PO Matrix

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	P012
CSDLO5011.1	3	3	2	-	-	-	-	-	-	-	-	3
CSDLO5011.2	3	3	3	-	-	-	-	-	-	-	-	3
CSDLO5011.3	3	3	3	3	-	-	-	-	-	-	-	3
CSDLO5011.4	2	2	3	3	-	-	-	-	-	-	-	3
CSDLO5011.5	2	3	3	3	-	-	-	-	-	-	-	3
CSDLO5011.6	2	3	3	3	-	-	-	-	1	1	-	3

Correlation Levels: 1: Slightly 2: Moderately 3: Substantially

Assessment Instruments

Course Outcome	Assessment – Direct	Maximum Marks	Assessment - Indirect
	Module Test - 1	10	Course Exit Survey
	Assignment 1	03	
	Internal Assessment I		
	Module Test - 2	10	
	Assignment 1	03	
	Internal Assessment I		
	Module Test - 3	10	
	Assignment 3	03	
	Internal Assessment I		
	Module Test - 4	10	
	Assignment 4	03	
	Internal Assessment II		
	Module Test - 5	10	
	Assignment 5	03	
	Internal Assessment II		
	Module Test - 6	10	
	Assignment 6	03	
	Internal Assessment II		
All Course Outcomes	University Examination	80	

Enrolled Student List

Roll No	IEN	Name
1	12216037	ADEPWAR KRISHNA RAJU
2	12216051	ANDRE KARTIK SANTOSH
3	12216010	AYARE PRACHI VILAS VINAYA
4	12216019	BALGUDE SHUBHAM JAYRAM SHUBHALI
5	12246016	BANDAL PRATHMESH NIVRUTTI SUNITA
6	12216030	BANSODE PRATHAM GANGADHAR
7	12246026	CHAVAN PRATHAMESH SACHIN
8	12246023	CHAVAN SAHIL SANJAY
9	12216045	DAHIBHAVKAR ATHARVA RAJESH
10	12216038	DATKHILE SIDDHESH KHANDU
11	12246005	GADE PRATHAMESH MAHADEO ARCHANA
12	12216033	GAWADE OM SANTOSH
13	12226007	GAWADE SIDDHESH SATISH
14	12216032	GHEWADE SHAILESH GANPAT
15	12246027	HARSHHINI HS HARI
16	12246020	JAISWAR TRISHNA ANIL
17	12216022	JOSHI RASIKA DEVENDRA
18	12216034	KHADE NIDHI RAVINDRA
19	12216005	KHAN JUMAN ASLAM HUMA
20	12216047	KHAN ZAID JAINUL ABIDIN
21	12246049	KUMBHAR HARSHADA DEVANAND
22	12216015	MADHAVI RUDRA ATUL KADAMBARI
23	12246038	MOGALREDDY SINDHU SRINIVAS
24	12246006	MUNDEWADI NIMIT SUBRAMANYA MEENAKSHI

25	12216043	PADMAWAR AAYUSH GAJANAN
26	12216021	PAL DAKSH DAYASHANKAR JAIDEVI
27	12216002	PANDEY NEHA PRADEEP REETA
28	12246036	PARAB OMKAR SHIVAJI
29	12216004	PATANE SHRAVANI SANTOSH LATA
30	12216016	PATARE ACHAL KAMALAKAR KALYANI
31	12216050	PATHARWAT TEJAS KAILASH
32	12226006	PATIL ARCHIT SANTOSH
33	12216028	PATIL CHINMAY ROHIT
34	12246025	PATIL DIYA SANTOSH
35	12216009	PATIL PARTH SHIVAJI SMITA
36	12216044	PATIL SANKET BHASKAR
37	12216013	RAHANGDALE DEEKSHANT LALIT ANITA
38	12246048	RAJBHAR VAISHANAVI VIPIN
39	12216011	SAID SAURAV RAJESH VAISHALI
40	12216039	SANGALE VINEET SHIVAJI
41	12216041	SATAM BHAVANA SATYAJEET
42	12226003	SATAV PRATHAMESH DEEPAK
43	12216025	SAWARDEKAR SUJAL MANGESH MAYURI
44	12216018	SHINDE PAVAN BALKRISHNA
45	12216024	SHIPURE PURVA MADHAV URMILA
46	12246045	TAWDE PIYUSH VINOD
47	12216029	VADKE YASH VINOD

Course Pre-requisite Survey

Prerequisite Questions

1) Find the median of the set = {2, 4, 4, 3, 8, 67, 23}*

8

2

3

4

2) Which of the following ways can be used to calculate median of an ordered list? Here, 'N' is the number of observations in an ordered list.
(Select all options which are applicable)

N+1

(N+1)/2 th value when N is odd

N/2 th value

((N/2th) value + (N/2th+1) value)/2 when N is even

3) Which of the following statement is Not true about skewness

It's a measure of dispersion

It's a measure of the shape of a distribution

It's a measure of Asymmetry

Tells the amount and direction of skew

4) Which of the following measures of central tendency is highly impacted by extreme values (outliers)?

Mode

Mean

Medium

None of the above

5) Suppose the average score on a national test is 500 with a standard deviation of 100. If each score is increased by 25, what would be the new mean and standard deviation?

500, 100

500, 125

525, 100

525, 105

6) Which of the following is true statements?

I. If the variance of a sample is zero, the variance of its population is also zero.

II. If the variance of the population is zero, the variance of a sample of the population would also be zero.

III. If the variance of a sample is zero, the sample mean and the sample median would be equal.

I and II

I and III

II and III

I, II and III

7) Compute the 'median' for the Marks obtained by the students in Python exam,

Marks: 81, 62, 85, 65, 76, 86, 78, 80, 87

76

85

78

80

8) Coefficient of Correlation ranges between

0 to 1

-1 to +1

0 to infinity

-infinity to +infinity

9) Can the "Measures of variability" be negative?

Measures of variability:- Range, IQR, Standard Deviation, and Variance

True

False

10) The coefficient of variation is given by

Standard deviation / Mean * 100

Standard deviation / Median * 100

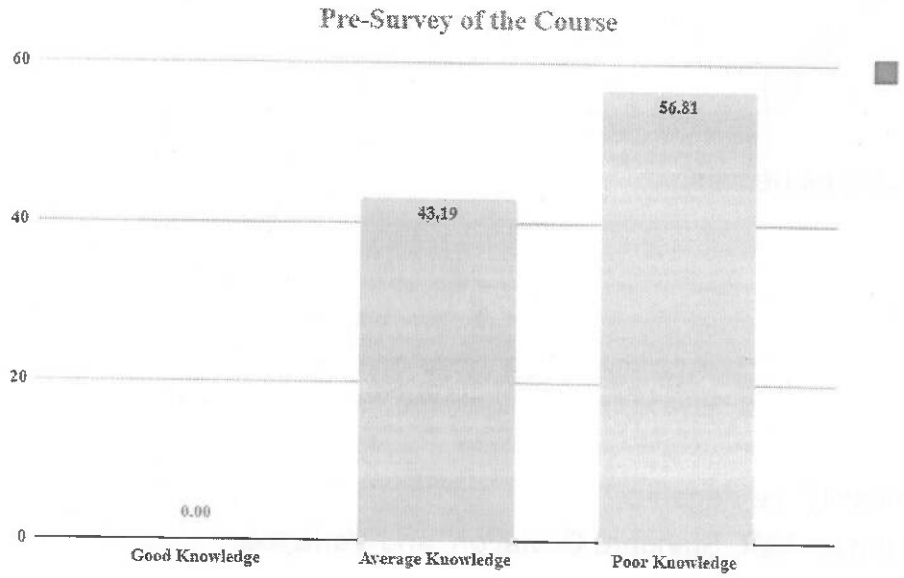
Mean / Standard deviation * 100

Median / Standard deviation * 100

Survey Analysis (from Annexure 1)

	Total (%)
Total	100.00
Good Knowledge	0.00
Average Knowledge	43.19
Poor Knowledge	56.81

Assessment Tool- Bar Graph



Assignments

Assignment No 1:

<table>
<tr><th>Sr.No.</th><th>Questions</th><th>CO:PO Mapping</th></tr>
<tr><td>1</td><td>What is the use of Scatter plot and Box plot</td><td>CSDLO5011.1: PO1, PO2, PO3, PO12</td></tr>
<tr><td>2</td><td>Calculate the first four moments of the following distribution about mean and hence
<table>
<tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr>
<tr><td>f</td><td>1</td><td>8</td><td>28</td><td>56</td><td>70</td><td>56</td><td>28</td><td>8</td><td>1</td></tr>
</table>
find β1 and β2.</td><td>CSDLO5011.1: PO1, PO2, PO3, PO12</td></tr>
<tr><td>3</td><td>The distribution of fortnightly wages of 280 employees of an undertaking is as given. Find the first quartile, the median and the third quartile. Find D4, P66,
<table>
<tr><td>Fortnightly Wages</td><td>Less than 200</td><td>200-400</td><td>400-600</td><td>600-800</td><td>800-1000</td><td>1000-1200</td><td>1200-1400</td><td>Above 1400</td></tr>
<tr><td>Frequency</td><td>12</td><td>16</td><td>38</td><td>78</td><td>80</td><td>35</td><td>14</td><td>7</td></tr>
</table>
P10, and P90.</td><td>CSDLO5011.1: PO1, PO2, PO3, PO12</td></tr>
<tr><td>4</td><td>Find Q1, Q2 and Q3 for the following dataset.
{2, 6, 7, 8, 8, 11, 12, 13, 14, 15, 22, 23}
Draw Box-and-Whisker plot. Identify outliers.</td><td>CSDLO5011.1: PO1, PO2, PO3, PO12</td></tr>
</table>

Assignment No 2:

Sr.No.	Questions	CO:PO Mapping
1	State and prove central limit theorem	CSDLO5011.2: PO1, PO2, PO3, PO12
2	Discuss Boot strapping vs. re-sampling	CSDLO5011.2: PO1, PO2, PO3, PO12
3	The mean yield for one-acre plot is 662kilos with a standard deviation 32 kilos. Assuming normal distribution, how many one-acre plots in a batch of 1000 plots would you expect to have yield i. Over 700 kilos ii. Below 650 kilos What is the lowest yield of the best 100 plots?	CSDLO5011.2: PO1, PO2, PO3, PO12
4	4. An agent sells life insurance policies to five equally aged, healthy people. According to recent data, the probability of a person living inthese conditions for 30 years or more is 2/3. Calculate the probability that after 30 years if i. All five people are still living. ii. At least three people are still living. iii. Exactly two people are still living.	CSDLO5011.2: PO1, PO2, PO3, PO12